

Evidence-based **drug repurposing candidates for chromosomal instability (CIN) syndromes like MVA1/BUB1B-related disease remain preclinical or early-phase;** none has been tested specifically in a child with biallelic BUB1B loss-of-function mutations, and standard COG/EpSSG [rhabdomyosarcoma](#) chemoradiotherapy protocols remain the evidence-based backbone of oncologic treatment.[\[1\]](#) The candidates below exploit the biology of spindle-assembly-checkpoint (SAC) deficiency and aneuploidy-driven proteotoxic/replication stress, but should be considered investigational, ideally within a clinical trial or compassionate-use framework with close multidisciplinary oversight given the patient's underlying genomic instability disorder.

Priori ty	Drug (generi c/brand)	Mechani sm of action	Target pathw ay	Clinical develop ment stage	Evidenc e in aneuploi d/cancer models	Pediatric safety profile	Contraindi cations/red flags	Key evidenc e gaps
1	Adavosertib (AZD1775; no approved brand)	WEE1 kinase inhibition → unrestrained CDK1 activation, forced mitotic entry with unrepaired damage, mitotic catastrophe	WEE1	Phase I/II trials in solid tumors, including combination with gemcitabine in HGSOc showing improved PFS [2]	Selective synthetic lethality demonstrated in aneuploid vs. diploid breast cancer models; rationale strengthened in high-CIN tumors [2] , [3]	No established pediatric dosing/safety database; adult trials show myelosuppression, diarrhea, fatigue	Combine cautiously with DNA-damaging chemotherapy (additive marrow toxicity); TP53 status effect on sensitivity is inconsistent [3]	No data in MVA1/BUB1B-mutant patients or pediatric RMS; optimal monotherapy vs. combination dosing in a SAC-deficient background undefined
2	Sovilnesib / VLS-1488 (KIF18A inhibitor),	Blocks kinesin KIF18A, exploiting reliance of chromosomally	Spindle/ mitotic motor (SAC-adjacent, "other")	Phase I complete for sovilnesib (NCT04293094, data pending)	Preclinical synthetic lethality specifically in chromosomally	No pediatric data; agent class is novel and unstudied	None established yet given early stage; theoretical concern for exacerbating mitotic	Mechanistically the most biologically aligned with

3	investigational)	unstable cells on this motor for mitotic spindle stabilization → selective mitotic death in CIN-high cells Inhibits HSP90–CDC37 chaperone complex, impairing folding/maturation of oncogenic kinases; aneuploid cells have insufficient chaperone capacity relative to their imbalanced proteome, producing selective toxicity	and ongoing (NCT06084416); VLS-1488 recruiting (NCT05902988)[2]	unstable tumor models[2]	d outside adult solid-tumor trials	errors in an already SAC-deficient cell population	SAC/CIN deficiency, but zero pediatric or MVA-specific data exist	
	Tanespimycin (17-AAG) and other HSP90 inhibitors	HSP90 (proteotoxic stress)	Tanespimycin reached but did not advance past phase I/II trials (poor pharmacokinetics, hepatotoxicity); no HSP90 inhibitor is FDA-approved[4], [5], [6]	Selective killing of aneuploid vs. euploid cells shown across independent model systems[7]	Not established in children; adult trials limited by dose-limiting hepatotoxicity and poor PK[5]	Hepatotoxicity, QT prolongation reported with some HSP90 inhibitors; caution combining with other hepatotoxic chemotherapy used in RMS regimens	No trials in MVA/CIN-syndrome patients; whether the aneuploidy-selective effect translates to clinically tolerable doses in a growing child is unknown	
4	IGF-1R–targeted agents (e.g., ganitumab, cixutum	Blocks IGF-1R signaling, a pathway relevant to rhabdom	IGF-1R	Prior pediatric/adolescent phase II trials in RMS and Ewing sarcoma;	Efficacy signal modest and largely negative in randomized	Some pediatric safety data exist from prior RMS/Ewing trials	Not disease-modifying for the underlying SAC defect; hyperglyce	No mechanistic link to aneuploidy-induced proteoto

umab; largely discontinued program(s)	yo sarcoma proliferation independent of CIN biology	several programs discontinued due to lack of survival benefit	zed pediatric sarcoma trials	(hyperglycemia, thrombocytopenia)	mia risk requires monitoring	xic stress; repurposing rationale here is RMS-histology-driven, not CIN-driven
5 (exploratory, low priority)	Senolytics (dasatinib + quercetin, navitoclax)	Selective clearance of senescent cells, which can accumulate after aneuploidy-induced stress and secrete pro-tumorigenic SASP factors	Early-phase adult trials in non-oncologic senescence-associated conditions; navitoclax has oncology development (BCL-2/BCL-XL inhibition) but not for CIN syndromes	No direct evidence identified in the aneuploidy/CIN-model literature reviewed here	Navitoclax causes dose-limiting thrombocytopenia; not studied in children with MVA	Rationale is extrapolated, not directly evidenced; risk of on-target marrow toxicity (navitoclax) compounded by chemotherapy-induced cytopenias
		Entirely theoretical for this indication; no aneuploidy-specific senolytic data were found				

Agents to flag as potentially problematic in this SAC-deficient background: standard spindle poisons (vincristine, vinblastine) remain part of guideline-based RMS therapy and should not be withheld, but there is a theoretical mechanistic concern that further SAC perturbation on top of germline BUB1B deficiency could amplify chromosome missegregation in surviving normal tissue; this has not been demonstrated clinically and is not a basis to alter standard-of-care dosing without trial data. Investigational HSP90 inhibitors with significant hepatotoxicity should be avoided if combined with hepatotoxic components of RMS chemotherapy (e.g., dactinomycin) outside a monitored trial setting.[\[5\]](#)

Overall, the WEE1 inhibitor adavosertib and KIF18A inhibitors have the strongest mechanistic and preclinical rationale for exploiting CIN/SAC deficiency specifically, while HSP90 inhibition has the longest-standing but clinically unproven rationale for aneuploidy-selective proteotoxicity.[\[2\]\[3\]\[7\]\[4\]](#) Given the absence of pediatric or MVA-specific trial data for any of these agents, use outside of standard rhabdomyosarcoma protocols should occur only through a clinical trial, compassionate-use pathway, or genomic tumor board with informed multidisciplinary and family discussion, alongside the cancer surveillance recommendations specific to MVA1 patients.[\[1\]](#)

Would you like to review the surveillance protocol recommended for children with MVA1/BUB1B-related cancer predisposition?

References

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